Monte Carlo Control without Exploring Starts

Reinforcement Learning

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1 The Problem with Exploring Starts

The previous report ended with Monte Carlo ES: a correct algorithm that resolves both fundamental problems of model-free control by working with $Q(s, a)$ and using exploring starts to guarantee that every (s, a) pair is visited infinitely often. The guarantee is real, but the assumption is not. Exploring starts requires that every episode can be initialised from an arbitrary state-action pair with positive probability. In any real physical system this fails immediately. A robot cannot be teleported to an arbitrary joint configuration; a game cannot be reset to an arbitrary mid-game state on demand.

The core issue is that exploring starts outsources the exploration requirement to the episode initialisation mechanism. What we need instead is a policy that guarantees coverage of all (s, a) pairs through its own action selection, at every time step, with no external assumptions. This report develops that idea.

2 Exploration Through the Policy Itself

2.1 Soft Policies

The minimal requirement for a policy to guarantee coverage is that no action is permanently excluded at any state.

Definition 2.1 (Soft policy)**.** *A policy π is* soft *if*

$$\pi(a \mid s) > 0 \qquad \forall\, a \in \mathcal{A}(s),\ \forall\, s \in \mathcal{S}.$$

A soft policy keeps every action reachable at every state at every time step. As long as the agent keeps acting, every (s, a) pair will eventually be visited, so $N(s, a) \to \infty$ without any special initialisation. Softness is the policy-level replacement for the exploring starts assumption.

2.2 ε-Soft Policies

Softness alone just says every probability is positive. It says nothing about how small those probabilities can be. In practice we want a concrete lower bound.

Definition 2.2 (ε-soft policy)**.** *A policy π is* ε-soft *if*

$$\pi(a \mid s) \geq \frac{\varepsilon}{|\mathcal{A}(s)|} \qquad \forall\, a \in \mathcal{A}(s),\ \forall\, s \in \mathcal{S},$$

for some fixed $\varepsilon > 0$.

The floor $\varepsilon/|\mathcal{A}(s)|$ is the uniform distribution scaled by ε: every action gets at least a $1/m$ share of the ε exploration budget. This is a quantitative version of softness. The remaining probability mass $1 - \varepsilon$ is free to be allocated however the policy chooses.

2.3 ε-Greedy Policies

Among all ε-soft policies, the most natural choice for control is the one that uses the free mass $1-\varepsilon$ as aggressively as possible: put all of it on the action with the currently highest Q-value.

The ε-greedy policy is most cleanly understood as a two-step sampling procedure. At each state s, flip a coin:

$$B \sim \text{Bernoulli}(\varepsilon).$$

- If $B = 1$ (with probability ε): select an action uniformly at random from all $m = |\mathcal{A}(s)|$ actions.
- If $B = 0$ (with probability $1 - \varepsilon$): select the greedy action $a^* = \arg\max_{a' \in \mathcal{A}(s)} Q(s, a')$.

We can read off the resulting distribution over actions directly. For the greedy action a^*, it is selected either because $B = 0$ or because $B = 1$ and the uniform draw happened to land on a^*:

$$\pi(a^* \mid s) = (1 - \varepsilon) \cdot 1 + \varepsilon \cdot \frac{1}{m} = 1 - \varepsilon + \frac{\varepsilon}{m}.$$

For any non-greedy action $a \neq a^*$, it is selected only through the random draw:

$$\pi(a \mid s) = (1 - \varepsilon) \cdot 0 + \varepsilon \cdot \frac{1}{m} = \frac{\varepsilon}{m}.$$

These two cases together give the definition.

Definition 2.3 (ε-greedy policy)**.** *Let $m := |\mathcal{A}(s)|$. Given action-value estimates Q, the ε-greedy policy assigns*

$$\pi(a \mid s) = \begin{cases} 1 - \varepsilon + \dfrac{\varepsilon}{m} & \text{if } a = \arg\max_{a' \in \mathcal{A}(s)} Q(s, a'), \\ \dfrac{\varepsilon}{m} & \text{otherwise.} \end{cases}$$

Every ε-greedy policy is ε-soft by construction, since even non-greedy actions receive $\varepsilon/m > 0$. But not every ε-soft policy is ε-greedy: an ε-soft policy may spread the free mass $1 - \varepsilon$ across multiple actions in any proportion.

> *ε-greedy is the most exploitation-aggressive member of the ε-soft family. It satisfies the exploration floor exactly and commits all remaining probability to the current best action. It is the greediest policy that still qualifies as ε-soft.*

Why this solves the exploring starts problem. Under an ε-greedy policy, every action at every state has probability at least $\varepsilon/m > 0$ at every time step. As long as the agent continues to act, every (s, a) pair accumulates visits and $N(s, a) \to \infty$ without any assumptions on episode initialisation. We will make this precise when we discuss convergence. For now, the key point is that exploration is no longer a property of how episodes begin; it is a built-in property of the policy itself.

3 Policy Improvement in the Model-Free Setting

3.1 The Role of Improvement in GPI

Introducing an ε-greedy policy resolves the exploration problem, but it only addresses the *evaluation* side of GPI. Evaluation alone, no matter how long we run it, only tells us the value of the current policy π. It does not move us toward π^*. The optimisation happens entirely in the improvement step.

To see why, recall what GPI is doing. Policy evaluation estimates q_π for the current π. Policy improvement then asks: given what we now know about q_π, is there a better policy? In the DP setting, the answer was given by the Bellman optimality operator

$$(Tv)(s) = \max_{a \in \mathcal{A}(s)} \left[r(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s,a)\, v(s') \right],$$

whose fixed point is v^*. The greedy improvement step

$$\pi'(s) = \arg\max_{a \in \mathcal{A}(s)} Q(s,a)$$

is the sample-based, model-free analogue of applying T: instead of maximising over the full model, we maximise over the estimated Q-values. Without this step, the algorithm has no mechanism to move from one policy to a better one. Running evaluation forever under a fixed π gives q_π, not q^*.

3.2 Why Improvement Still Works with Approximate Estimates

The natural concern is: if our Q estimates are noisy because they come from a finite number of sampled trajectories, does the improvement step still make progress? After all, the Policy Improvement Theorem guarantees $v_{\pi'} \geq v_\pi$ only when $q_\pi(s, \pi') \geq v_\pi(s)$ for all s, and that antecedent was proved assuming exact values.

The answer has two parts.

First, even with approximate Q, the improvement step tends to move probability mass toward actions that appear better based on the data seen so far. This may not be a guaranteed improvement at every single step, but it is a systematic bias in the right direction.

Second, and more importantly, the improvement step is what creates the feedback loop that drives learning. Without it, the policy is fixed and evaluation is just estimating a constant. With it, a better Q estimate leads to a better policy, which generates better trajectories, which give better Q estimates. The GPI loop is the engine; the improvement step is the part that turns better estimates into better behaviour.

> *Policy evaluation alone answers the question: how good is my current policy? Policy improvement answers the question: given that, how should I act differently? Without improvement, GPI is just estimation. Without evaluation, improvement is blind. The two steps must alternate for learning to happen.*

3.3 Connection to the Policy Improvement Theorem

The formal justification for the improvement step is the ε-Greedy Policy Improvement Theorem, which we state here and prove in the next section.

The theorem says: if π is any ε-soft policy and π' is the ε-greedy policy derived from q_π, then

$$v_{\pi'}(s) \geq v_\pi(s) \qquad \forall\, s \in \mathcal{S}.$$

This is the ε-soft analogue of the standard Policy Improvement Theorem. It guarantees that within the class of ε-soft policies, the ε-greedy update never makes the policy worse. The proof does not

assume exact Q-values; it assumes exact q_π values. In the Monte Carlo setting, evaluation runs long enough that $Q \approx q_\pi$ closely, and improvement is applied to those estimates. As evaluation improves, the quality of the improvement step improves with it.

4 ε-Greedy Policy Improvement Theorem

4.1 Setup

We already proved the Policy Improvement Theorem in a previous report: if

$$q_\pi(s, \pi') \geq v_\pi(s) \qquad \forall\, s \in \mathcal{S},$$

then $v_{\pi'}(s) \geq v_\pi(s)$ for all s. We therefore only need to establish the antecedent for the ε-greedy update starting from an ε-soft π.

Theorem 4.1 (ε-Greedy Policy Improvement)**.** *Let π be any ε-soft policy. Let π' be the ε-greedy policy with respect to q_π. Then*

$$v_{\pi'}(s) \geq v_\pi(s) \qquad \forall\, s \in \mathcal{S}.$$

Proof. Fix any $s \in \mathcal{S}$. Let $m := |\mathcal{A}(s)|$ and let $a^* \in \arg\max_{a \in \mathcal{A}(s)} q_\pi(s, a)$.

Step 1: Expand $q_\pi(s, \pi')$.

The ε-greedy policy π' splits its mass into a uniform floor of ε/m on every action and the entire remaining mass $1 - \varepsilon$ on a^*:

$$\begin{aligned} q_\pi(s, \pi') &= \sum_{a \in \mathcal{A}(s)} \pi'(a \mid s)\, q_\pi(s, a) \\ &= \underbrace{\frac{\varepsilon}{m} \sum_{a \in \mathcal{A}(s)} q_\pi(s, a)}_{\text{mandatory floor}} + \underbrace{(1 - \varepsilon)\, q_\pi(s, a^*)}_{\text{free mass on } a^*}. \end{aligned}$$

Since a^* is the maximising action,

$$q_\pi(s, \pi') = \frac{\varepsilon}{m} \sum_{a \in \mathcal{A}(s)} q_\pi(s, a) + (1 - \varepsilon) \max_{a \in \mathcal{A}(s)} q_\pi(s, a). \tag{1}$$

Step 2: Apply the same decomposition to $v_\pi(s)$.

Write $v_\pi(s) = \sum_a \pi(a \mid s)\, q_\pi(s, a)$ and pull the mandatory floor from each $\pi(a \mid s)$. Since π is ε-soft, $\pi(a \mid s) - \varepsilon/m \geq 0$ for all a:

$$v_\pi(s) = \underbrace{\frac{\varepsilon}{m} \sum_a q_\pi(s, a)}_{\text{same mandatory floor as (1)}} + \underbrace{\sum_a \left(\pi(a \mid s) - \frac{\varepsilon}{m} \right) q_\pi(s, a)}_{\text{how } \pi \text{ uses its free mass } (1-\varepsilon)}.$$

The first terms of $v_\pi(s)$ and $q_\pi(s, \pi')$ are identical. The two policies differ only in how they spend the free mass $1 - \varepsilon$.

Step 3: The free mass under π' dominates that under π.

The residual weights $\pi(a \mid s) - \varepsilon/m$ are nonnegative and sum to $1-\varepsilon$. Normalising,

$$w(a) := \frac{\pi(a \mid s) - \varepsilon/m}{1-\varepsilon} \geq 0, \qquad \sum_a w(a) = 1,$$

so w is a probability distribution over $\mathcal{A}(s)$. The free-mass term of $v_\pi(s)$ is therefore a convex combination of $q_\pi(s,a)$ values scaled by $1-\varepsilon$. A convex combination never exceeds the maximum:

$$\sum_a \left(\pi(a \mid s) - \frac{\varepsilon}{m}\right) q_\pi(s,a) = (1-\varepsilon) \sum_a w(a)\, q_\pi(s,a) \;\leq\; (1-\varepsilon) \max_a q_\pi(s,a).$$

Step 4: Conclude.

The right side above is the free-mass term of $q_\pi(s,\pi')$ in (1), and the mandatory-floor terms cancel. Therefore:

$$v_\pi(s) = \frac{\varepsilon}{m} \sum_a q_\pi(s,a) + (1-\varepsilon) \sum_a w(a)\, q_\pi(s,a) \;\leq\; \frac{\varepsilon}{m} \sum_a q_\pi(s,a) + (1-\varepsilon) \max_a q_\pi(s,a) = q_\pi(s,\pi').$$

So $q_\pi(s,\pi') \geq v_\pi(s)$ for all s. By the Policy Improvement Theorem,

$$v_{\pi'}(s) \geq v_\pi(s) \qquad \forall\, s \in \mathcal{S}. \qquad \square$$

$\square$

Every ε-soft policy is forced to spend probability ε/m on every action. That part is identical across all ε-soft policies and cancels in the comparison. The only freedom is in how the remaining mass $1-\varepsilon$ is distributed. The old policy π spreads it arbitrarily; ε-greedy puts all of it on the best action. Concentrating mass on the maximum can never decrease the weighted average. That is the entire argument.

5 GLIE Monte Carlo Control

5.1 The Cost of a Fixed ε

The ε-Greedy Policy Improvement Theorem guarantees that each improvement step never degrades the policy within the ε-soft class. But a fixed $\varepsilon > 0$ means the policy can never become fully greedy: every non-greedy action retains probability ε/m forever. The algorithm converges, but to the best ε-soft policy, not to π^*. To reach π^*, ε must decay to zero, but slowly enough that every (s,a) pair is still visited infinitely often. This tradeoff is captured by the GLIE condition.

5.2 GLIE

Definition 5.1 (Greedy in the Limit with Infinite Exploration (GLIE))**.** *A sequence of policies $\{\pi_k\}_{k\geq 1}$ satisfies GLIE if:*

(1) All state-action pairs are explored infinitely often:

$$\lim_{k\to\infty} N_k(s,a) = \infty \qquad \forall\, (s,a).$$

(2) The policy converges to the greedy policy:

$$\lim_{k\to\infty} \pi_k(a \mid s) = \mathbf{1}\left\{a = \arg\max_{a'\in\mathcal{A}(s)} Q_k(s,a')\right\} \qquad \forall\,(s,a).$$

Condition (1) ensures every $Q(s,a)$ estimate has enough data to converge. Condition (2) ensures that as the estimates converge, the policy converges to the greedy policy with respect to those estimates, which must then be π^*.

The canonical schedule satisfying both conditions is $\varepsilon_k = 1/k$. Condition (1) holds because $\varepsilon_k > 0$ for all finite k, so every action remains reachable throughout training. Condition (2) holds because $\varepsilon_k \to 0$, so the probability of the greedy action converges to 1.

5.3 Algorithm and Convergence

Key Result

GLIE Monte Carlo Control.
Initialise $Q(s,a)$ arbitrarily and $N(s,a) = 0$ for all (s,a).
For each episode $k = 1, 2, 3, \ldots$**:**

1. Sample episode $\{S_1, A_1, R_2, \ldots, S_T\} \sim \pi_k$.
2. For each (S_t, A_t) in the episode, compute G_t and update:

$$N(S_t, A_t) \leftarrow N(S_t, A_t) + 1,$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{1}{N(S_t, A_t)}\big(G_t - Q(S_t, A_t)\big).$$

3. Decay ε and improve the policy:

$$\varepsilon \leftarrow \frac{1}{k}, \qquad \pi_{k+1} \leftarrow \varepsilon\text{-greedy}(Q).$$

Theorem 5.1 (Convergence of GLIE Monte Carlo Control)**.** *GLIE Monte Carlo control converges to the optimal action-value function:*

$$Q(s,a) \to q_*(s,a) \qquad \text{as } k \to \infty.$$

The two GLIE conditions do exactly the work needed. Condition (1) guarantees the Monte Carlo estimator is consistent at every (s,a). Condition (2) guarantees the policy the estimator converges to is actually greedy with respect to the limiting Q, which must therefore satisfy the Bellman optimality equations and equal q^*. We defer the formal proof to a later report.